**Technical Appendix: A Structure-Provenance Framework for Synthetic Organizations**

Supplement to the Stage 1 proposal "Emergence or Artifact? A Structure-Provenance Framework for Synthetic Organizations". Authors withheld for review. Version of 27 October 2026.

## Overview

This appendix specifies everything the proposal's numbers rest on, so that a reader can inspect the framework and reproduce every reported value. It gives the estimators (A1), the null models and their assumptions (A2), the synthetic environment and conditions (A3), the full known-answer test (A4), the plan-specified LLM comparisons with their sensitivity analyses (A5), the exploratory LLM conditions (A6), the empirical reference cases (A7), a sample provenance report (A8) and reproduction instructions with a clean-environment check (A9).

All code and stored runs are in the anonymized repository: anonymous.4open.science/r/synthetic-org-benchmark-45E2. Paths below are relative to its `code/` folder.

*Table 1*. Where each component is in the repository

| What | Where in the repository |
|---|---|
| Estimators and nulls | `diffenv/measures.py` |
| One-command provenance report | `provenance_report.py`, `examples/` |
| Environment, agents, label shuffles | `diffenv/env.py`, `diffenv/policies.py`, `diffenv/run.py` |
| Known-answer test | `diffenv/ground_truth.py`; results in `diffenv_results/ground_truth_2026-10-17b/` |
| Analysis plan and its scoring | `ANALYSIS_PLAN_2026-10-16.md`, `score_plan.py`, `score_plan_sensitivity.py`; results in `diffenv_results/plan_scores_2026-10-17/` |
| Every LLM run (event log, prompts, every raw reply, tokens, cost) | `diffenv_results/{final, alias_rerun, gaps_2026-10-16, replications_2026-10-17, balanced_2026-10-17}/` |
| Reference cases | `diffenv/run_substrates.py`, `diffenv/run_external.py`; data sources in `../DATA.md` |
| Figure 1 | `figures/make_figure1_v4.py` |

## A1. Persistence estimators

The three repeatability-based dimensions (individual, functional, structural) are scored by one function, `measures.differentiation()`, on a long table of rows (run, window, member, value). Nothing in it knows whether the rows come from ants, offices or agents. The functional dimension applies it to each task type's shares (`measures.specialization()`). The relational dimension is different: `measures.partner_consistency()` scores the Jaccard overlap of each member's partner sets in consecutive windows against random partner sets of the same sizes (A2), not repeatability against a label permutation.

**Standardization.** Within each run and window w, values are z-scored across the members present, using the sample standard deviation. A window with fewer than two members is dropped; a window in which all members tie is set to zero and counted as degenerate, never silently removed.

$$z_{iw} = \frac{y_{iw} - \bar{y}_w}{s_w}$$

**Repeatability.** For each member i with at least two windows, let z̄_i be the mean of its standardized scores. Repeatability is the share of variance that lies between members (Nakagawa & Schielzeth 2010), with sample variances (ddof = 1) throughout:

$$R = \frac{V_B}{V_B + V_W}, \qquad V_B = \mathrm{Var}_i\,(\bar{z}_i), \qquad V_W = \frac{1}{n}\sum_i \mathrm{Var}_w\,(z_{iw})$$

V_B is not corrected for the sampling variance of member means, so R is positive even when members are exchangeable: with 9 members and 10 windows its null median is about 0.09 and its 95th percentile about 0.18. That is why R is never reported without its own baseline (A2).

**Split-half reliability** is the Spearman correlation between each member's mean in the first and second half of its windows, with the same null.

**The four dimensions.** Each is a choice of the value y and is scored with R unless stated.

*Table 2.* The four dimensions of differentiation

| Dimension | Value per member and window | Notes |
|---|---|---|
| Individual | Work actions per round in which the member had at least one workable item | Opportunity adjustment: windows with no opportunity are dropped. Raw work counts are also reported |
| Functional (favored-type R) | Share of the member's work on each task type t, for member-windows with at least 2 work actions | R is computed for each type with at least 3 members and 2 windows, then averaged over types; the baseline is the mean of the per-type 95th percentiles. Secondary: within-window entropy of task types against a type-shuffle null |
| Relational (partner consistency) | Jaccard overlap of a member's interaction partners in consecutive windows | Mean over consecutive pairs where both sets are non-empty; compared with random partner sets (A2), not with R |
| Structural (hub persistence) | Interactions received per window | In the LLM environment, "tells" received |

**Signatures used for attribution.** These need no access to the agents' internals.

- *Growth:* task-type R in rounds 1–50 against rounds 51–100.
- *Continuity:* the share of agents whose most-worked type is the same in rounds 1–50 and 51–100, against a null that permutes second-half favorites across agents. This is the read-out of the history wipe.

- *Recurrence across runs:* how often the same agent has the same favorite type (chance 1/4 with four types), or is the top receiver, across independent runs.

**Decision rule.** A run is above baseline when its observed value exceeds the 95th percentile of its own null (200 draws, fixed seed). For the functional dimension, whose statistic is a mean over task types, the pre-specified threshold is the mean of the per-type 95th percentiles. The 5% expectation below holds exactly only for a statistic compared with its own null distribution, so A5 also reports the comparison with the 95th percentile of the null of the mean itself; the pre-specified threshold turns out to be the more conservative of the two. Across runs we report the count above baseline against the 5% expected by chance; with 10 runs, 3 or more above baseline has binomial probability 0.012 under the null. Conditions are summarized by the median and a bootstrap 95% interval for the median (10,000 resamples of runs).

### A2. Null models and their assumptions

Each statistic has its own reference distribution, built by destroying member identity while keeping everything else the statistic depends on. The table states what each null preserves and what it assumes.

*Table 3.* Null models: how each draw is made, what it preserves and what it assumes

| Null | Used for | How a draw is made | Preserves | Assumes |
|---|---|---|---|---|
| Label permutation | R and split-half on the repeatability-based dimensions (individual, functional, structural); partner consistency has its own null below | Shuffle values among the members present, independently within each run and window; 200 draws | Every value, each window's distribution, group-level trends and shocks over time, who is present when | Members are exchangeable within a window: under no differentiation, any member could have produced any value in that window |
| Type shuffle | Within-window entropy (functional, secondary) | Shuffle task-type labels among all work actions in a window | How much each member works; how common each type is in each window | Task types are equally available to every member within a window |
| Random partner sets | Partner consistency (relational) | Replace each member's partners in two consecutive windows with random sets of the same sizes drawn from the other members | Each member's number of partners per window | Any colleague is an equally available partner |
| Favorite permutation | Continuity across the midpoint (known-answer test) | Permute second-half favorite types across agents | The distribution of favorite types in each half | Favorites in the two halves are unrelated under no history dependence |
| Chance recurrence | Same favorite type, same top receiver across runs | Analytic: 1/4 for four equally used types; 1/9 per run for nine agents | — | Agent identities are comparable across runs (same index, same environment seed) |

**Where exchangeability can fail, and what the framework does.** Members may differ in opportunity: under private visibility, each arriving item is revealed to one agent, so some agents see more work. The individual measure therefore scores work per round with opportunity, and the known-answer test reports opportunity per agent-round for every condition (A4). Stage 2 adds a baseline that permutes values only among members with similar exposure, a partner baseline that holds label positions fixed (A5 shows why), and a block-permutation baseline robust to dependence across adjacent windows.

**Inference across runs.** Runs are independent replications with different seeds. Plan-specified contrasts use exact two-sided Mann–Whitney tests on run-level R, Holm-corrected across the four contrasts. Sensitivity analyses use exact Wilcoxon signed-rank tests paired by seed, because each seed starts the same environment in every condition before behavior diverges.

### A3. Synthetic environment, agent conditions and label shuffles

Nine agents share an open queue in which no one is in charge, no one has a role and there is no individual score; whether they come to differ is the measurement (`diffenv/env.py`).

**Rules.** Each round, 6 items arrive, each of one of 4 types (alpha, beta, gamma, delta). An item completes when two different agents have worked it and expires 5 rounds after arrival. Under private visibility, used for every reported LLM run, each arriving item is revealed to exactly one agent, so a second worker can only come through a `tell`. Each round every agent takes one action: `work <item>`, `tell <person> <item>` or `pass`; decisions within a round are simultaneous. A run lasts 100 rounds, scored in ten windows of ten rounds. Each agent sees the items it knows of, in random order, messages received that round, and its own last five actions. The collective payoff is not shown during the run.

**Model settings.** gpt-4.1-mini (gpt-4.1-mini-2025-04-14) and Claude Haiku 4.5 (claude-haiku-4-5-20251001), temperature 1.0, at most 40 output tokens, one call per agent per round (900 calls per run). A reply that cannot be parsed is recorded and treated as `pass`; none of the 60 runs in the plan-specified comparisons had an unparseable reply, and the exploratory runs had at most 4 in 2,700 calls. Every call is journaled, so an interrupted run resumes without repeating a call, and hard caps on calls and cost are checked before each call.

**The prompt every agent receives (condition C1), verbatim:**

> You are one member of a small team of 9 people working through a shared queue of items. Nobody is in charge and nobody has a role. Each item has a type (alpha, beta, gamma, delta) and is finished only after two different people have worked on it. Items disappear if nobody finishes them within a few rounds. You only ever see a few of the open items; other people can tell you about items you cannot see, and you can tell them. The team's goal is to finish as many items as possible. There is no individual credit.

The prompt then lists the three actions and asks for the action alone, one line, no explanation. Each round's message gives the round, the agent's own name, the other members' names, messages received, the items it could work, items it has already worked, and its recent actions.

*Table 4.* Agent conditions in the synthetic environment

| Condition | Agents | What differs from C1 |
|---|---|---|
| C0 | Rule-based, identical: work a random known item with probability 0.5, else tell a random colleague about one | Negative control; must sit at the baseline |
| C0+ | Rule-based, each with a fixed favorite type and a work propensity spread from 0.2 to 0.9 | Positive control; must be detected |
| C1 | LLM, identical prompt | Nothing assigned |
| C2 | LLM, told a specialty | Adds: "Your specialty is items of type {t}. Prioritise working on {t} items and tell others about items of the other types." Types assigned in rotation by agent index |
| C3 | LLM, given a disposition | Adds one of nine one-sentence dispositions (cautious, proactive, sociable, independent, conscientious, easy-going, curious, helpful, focused); no functional role |
| C1 + competence | LLM, identical prompt | Hidden per-agent, per-type success probabilities; a failed `work` does not count |
| C1 + exit | LLM, identical prompt | Adds a permanent `leave` action |

**Label shuffles.** By default an agent sees its colleagues under fixed names A0–A8. Two per-viewer relabelings test whether structure follows names. Both are fixed for the whole run, so a viewer's "A1" is always the same colleague and genuine partnerships could still form.

- *Latin square* (`--alias`): with a seeded permutation a of 0..8 and a seeded order of the nine names, viewer v sees agent x under the name with index (a[x] + a[v]) mod 9. Every viewer sees nine distinct names and every agent carries every name once, but each viewer's own name differs, so the agent whose own name is "A1" is no one's "A1".
- *Balanced* (`--alias-balanced`): with a seeded position pos, viewer v sees agent x as A{(pos[x] − pos[v]) mod 9}. Every viewer is A0 to itself, and every agent carries each colleague label, A1 included, for exactly one viewer. The test suite checks this property (`diffenv/test_alias.py`).

### A4. Known-answer test

The framework identifies the known source of differentiation in each of ten conditions run on rule-based agents in the same environment, with the same visibility and the same scoring as the LLM runs (`diffenv/ground_truth.py`, 20 runs per condition, seeds 0–19).

**Rule.** Every agent uses one response-threshold rule (Theraulaz, Bonabeau & Deneubourg 1998). For a known item of type t, the agent works it with probability

$$P(\text{work} \mid t) = \frac{1}{1 + \theta_t^2}$$

An agent considers its known items in random order and works the first one it accepts; if it accepts none, it tells a random colleague about a random item it knows (a `tell`). Conditions differ only in where the thresholds come from.

- *No source:* all thresholds fixed at 1.
- *Prescribed:* each agent fixed low (0.2) on one type and high (3.0) on the others, by agent index; *reassigned at random*: the same, with the assignment redrawn each run.
- *Acquired:* all start at 1; working type t lowers its threshold by 0.3 and raises the others by 0.1, within [0.2, 3.0].
- *History wiped:* at round 50 every agent's thresholds return to its starting values.
- *No communication:* acquired thresholds, but agents never tell. Under private visibility no item can then be completed; the condition isolates individual experience from any coupling between agents.
- *Label preference:* identical agents work a random known item with probability 0.5; otherwise they tell the colleague they see as "A1" with probability 0.6, else a random colleague; run with fixed names, the Latin-square shuffle and the balanced shuffle (A3).

*Table 5.* Known-answer test: results by condition (20 runs each)

| Condition | Task type | Hub | Work given opportunity | Task type, rounds 1–50 / 51–100 | Continuity across round 50 (chance) | Same favorite type across runs (chance 0.25) | Same top receiver across runs |
|---|---|---|---|---|---|---|---|
| No source | 0.09 (0.19), 0/20 | 0.08 (0.19), 2/20 | 0.10 (0.18), 1/20 | 0.18 / 0.17, 0/20 / 0/20 | 0.23 (0.25) | 0.26 | 4/20 |
| Prescribed | 0.49 (0.20), 20/20 | 0.09 (0.19), 0/20 | 0.10 (0.18), 2/20 | 0.57 / 0.50, 20/20 / 20/20 | 0.96 (0.26) | 1.00 | 4/20 |
| Prescribed, reassigned at random | 0.55 (0.19), 20/20 | 0.12 (0.19), 0/20 | 0.11 (0.18), 0/20 | 0.57 / 0.59, 20/20 / 19/20 | 0.96 (0.26) | 0.24 | 4/20 |
| Prescribed, reassigned, history wiped | 0.55 (0.19), 20/20 | 0.12 (0.19), 0/20 | 0.11 (0.18), 0/20 | 0.57 / 0.59, 20/20 / 19/20 | 0.96 (0.26) | 0.24 | 4/20 |
| Acquired from own history | 0.35 (0.19), 20/20 | 0.10 (0.19), 0/20 | 0.10 (0.19), 0/20 | 0.36 / 0.45, 12/20 / 19/20 | 0.43 (0.26) | 0.23 | 4/20 |
| Acquired, history wiped | 0.19 (0.19), 8/20 | 0.09 (0.19), 0/20 | 0.07 (0.19), 1/20 | 0.36 / 0.39, 12/20 / 15/20 | 0.24 (0.25) | 0.25 | 5/20 |
| Acquired, no communication | 0.25 (0.19), 18/20 | — | 0.09 (0.19), 1/20 | 0.30 / 0.32, 7/20 / 8/20 | 0.39 (0.26) | 0.24 | — |
| Label preference, fixed names | 0.09 (0.19), 0/20 | 0.97 (0.19), 20/20 | 0.09 (0.18), 1/20 | 0.18 / 0.18, 0/20 / 0/20 | 0.23 (0.24) | 0.26 | 20/20 |
| Label preference, Latin-square shuffle | 0.09 (0.19), 0/20 | 0.32 (0.19), 20/20 | 0.06 (0.19), 0/20 | 0.18 / 0.16, 0/20 / 0/20 | 0.18 (0.24) | 0.25 | 4/20 |
| Label preference, balanced shuffle | 0.10 (0.19), 1/20 | 0.10 (0.19), 0/20 | 0.08 (0.19), 1/20 | 0.19 / 0.18, 0/20 / 0/20 | 0.25 (0.25) | 0.25 | 5/20 |

Cells give median R (median baseline) and runs above baseline out of 20. Half-run baselines are about 0.34, because five windows give a noisier R. "Same top receiver" is the largest number of runs in which one and the same agent received the most tells.

**Contrasts** (bootstrap 95% intervals over runs):

- Continuity, acquired minus acquired-and-wiped, paired by seed: +0.19 [+0.10, +0.28]. The wipe removes history-dependent continuity.
- Continuity, prescribed-reassigned minus prescribed-reassigned-and-wiped: +0.00 [+0.00, +0.00]. The wipe leaves prescribed roles intact.
- Task-type R, communication minus no communication: +0.09 [+0.06, +0.13]. Communication also raises exposure: items an agent could work per round were 2.36 with communication and 1.78 without, so this contrast does not isolate interaction. Stage 2 (D2) holds exposure equal.

**How to read it.** Three readings bear on attribution. Prescribed and acquired specialization are both detected (20/20) but differ in signature: prescribed is stable from the start (0.57 then 0.50) and recurs on the same agent in every run when assignment is fixed (1.00), while acquired specialization grows (0.36 then 0.45) and lands on agents at chance (0.23). Recurrence alone does not discriminate, because prescribed roles reassigned at random also land at chance (0.24); the wipe does, because it changes histories while leaving assigned thresholds intact. The label rows show that a name preference produces a near-perfect hub, that the Latin-square shuffle leaves a residual hub in every run, and that only the balanced shuffle removes it. In the history-wiped condition, 8/20 runs stay above the whole-run task-type baseline (15/20 against the aggregate-null threshold of A5) because specialization re-forms after round 50; attribution rests on continuity across the wipe, not on the whole-run count.

**Opportunity.** Items an agent could work per round (median over runs): no source 2.18; prescribed 3.03; acquired 2.36; acquired, wiped 2.27; no communication 1.78; prescribed, reassigned 3.00; label preference 3.09–3.11.

### A5. Plan-specified LLM comparisons

All three comparisons came out as the plan's decision rules required, on the full ten-run cells, on the runs made after the plan alone, and paired by seed.

**The plan.** `ANALYSIS_PLAN_2026-10-16.md` was written on 16 October 2026 at 22:25 UTC. The replication runs had started at about 22:13 UTC; by then only journal line counts (calls completed) had been read, and no outcome of a new run had been inspected. The balanced-shuffle runs had not started. The plan fixed two primary outcomes (hub persistence, task-type persistence), the scoring code (unchanged since 30 September), and these rules:

- **C1.** gpt-4.1-mini, identical agents, fixed names vs the Latin-square shuffle, seeds 0–9. The artifact replicates if at least 8 of 10 fixed-name runs are above baseline on hub persistence and the agent named "A1" is the top receiver in at least 8; the shuffle removes it if at most 2 of 10 shuffled runs are above baseline.
- **C2.** The same fixed-name runs vs the balanced shuffle, seeds 0–9; same rules.
- **C3.** For each model, identical agents (Latin-square shuffle) vs agents told a specialty, seeds 0–9, task-type persistence. The contrast holds if at least 8 of 10 specialty runs and at most 2 of 10 identical-agent runs are above baseline.

- Two-sided Mann–Whitney tests on run-level R, Holm-corrected across the four tests; all other diagnostics reported as counts above baseline and labeled exploratory.

Seeds 0–2 of the gpt-4.1-mini fixed-name and Latin-square cells and of the Haiku Latin-square cell, and seeds 0–1 of the specialty cells, had been run before the plan; the plan included them by design. Their contribution is checked by the post-plan sensitivity analysis below.

**Results** (`score_plan.py`; `diffenv_results/plan_scores_2026-10-17/SCORES.md`).

*Table 6.* Plan-specified LLM comparisons: results by cell

| Cell | Outcome | Runs above baseline | Median R [bootstrap 95% CI] | Range | Median baseline |
|---|---|---|---|---|---|
| gpt-4.1-mini, fixed names | Hub | 10/10 | 0.36 [0.32, 0.40] | 0.26–0.50 | 0.18 |
| gpt-4.1-mini, Latin-square shuffle | Hub | 2/10 | 0.15 [0.10, 0.18] | 0.07–0.22 | 0.18 |
| gpt-4.1-mini, balanced shuffle | Hub | 0/10 | 0.08 [0.05, 0.10] | 0.01–0.11 | 0.18 |
| gpt-4.1-mini, identical (Latin-square) | Task type | 0/10 | 0.10 [0.07, 0.12] | 0.05–0.15 | 0.19 |
| gpt-4.1-mini, told a specialty | Task type | 10/10 | 0.71 [0.64, 0.74] | 0.60–0.78 | 0.21 |
| Haiku 4.5, identical (Latin-square) | Task type | 0/10 | 0.10 [0.07, 0.12] | 0.05–0.13 | 0.19 |
| Haiku 4.5, told a specialty | Task type | 10/10 | 0.63 [0.61, 0.66] | 0.59–0.70 | 0.20 |

The agent named "A1" was the top receiver in 8 of 10 fixed-name runs (A4 in the other two). Verdicts: C1, the artifact replicates and the Latin-square shuffle removes it by the plan's rule; C2, the balanced shuffle removes it in every run; C3, the contrast holds on both models. In all four tests U = 100 (complete separation), exact p = 0.00001, Holm p = 0.00004.

**Sensitivity analyses** (`score_plan_sensitivity.py`; `SENSITIVITY.md`).

*Table 7.* Sensitivity analyses of the plan-specified contrasts

| Contrast | Post-plan runs only: above baseline | Exact Mann–Whitney p (Holm) | Paired by seed: first higher | Exact Wilcoxon p (Holm) |
|---|---|---|---|---|
| C1: fixed vs Latin-square, hub | 7/7 vs 1/7 | 0.0006 (0.0009) | 10/10 | 0.002 (0.008) |
| C2: fixed vs balanced, hub | 7/7 vs 0/10 | 0.0001 (0.0004) | 10/10 | 0.002 (0.008) |
| C3: gpt-4.1-mini, specialty vs identical | 8/8 vs 0/7 | 0.0003 (0.0009) | 10/10 | 0.002 (0.008) |
| C3: Haiku 4.5, specialty vs identical | 8/8 vs 0/7 | 0.0003 (0.0009) | 10/10 | 0.002 (0.008) |

"A1" was the top receiver in 6 of the 7 post-plan fixed-name runs.

**Calibration of the functional baseline** (`functional_null_sensitivity.py`; `FUNCTIONAL_NULL_SENSITIVITY.md`). The pre-specified functional threshold is the mean of the per-type 95th percentiles (A1), which is not in general the 95th percentile of the null distribution of the mean statistic. We therefore also estimated that distribution directly: in each of 200 draws, one permutation of member labels per window is applied to all task types jointly, so each member-window keeps its whole type profile, and the mean of the per-type R is recomputed; the threshold is the 95th percentile of these draws. This is a sensitivity analysis. The pre-specified analysis, its thresholds and the verdicts above are unchanged, and the recomputed observed statistic matches the stored one exactly.

The aggregate-null threshold is lower (median about 0.15 against 0.19), so the pre-specified threshold is conservative. Against it, 1 of 20 identical-agent runs in the C3 cells is above baseline (gpt-4.1-mini, seed 0, a pre-plan run: 0.152 against 0.143), within the plan's limit of 2 of 10 per model; all 20 specialty runs remain above. Pooled over every identical-agent LLM run, 2 of 43 are above, close to the 5% expected by chance. In the known-answer test, the no-source condition has 1 of 20 above, prescribed and acquired specialization are detected in every run, and the history-wiped condition rises from 8 to 15 of 20 (see A4).

*Table 8.* Functional baseline: pre-specified and aggregate-null thresholds

| Cell | Runs | Median R | Pre-specified threshold (median) | Above | Aggregate-null threshold (median) | Above |
|---|---|---|---|---|---|---|
| gpt-4.1-mini, fixed names | 10 | 0.11 | 0.19 | 0/10 | 0.15 | 0/10 |
| gpt-4.1-mini, identical (Latin-square) | 10 | 0.10 | 0.19 | 0/10 | 0.14 | 1/10 |
| gpt-4.1-mini, balanced shuffle | 10 | 0.09 | 0.19 | 0/10 | 0.15 | 1/10 |
| gpt-4.1-mini, told a specialty | 10 | 0.71 | 0.21 | 10/10 | 0.16 | 10/10 |
| Haiku 4.5, identical (Latin-square) | 10 | 0.10 | 0.19 | 0/10 | 0.15 | 0/10 |
| Haiku 4.5, told a specialty | 10 | 0.63 | 0.19 | 10/10 | 0.15 | 10/10 |
| Haiku 4.5, fixed names (exploratory) | 3 | 0.09 | 0.19 | 0/3 | 0.15 | 0/3 |
| Rule-based: no source | 20 | 0.09 | 0.19 | 0/20 | 0.15 | 1/20 |
| Rule-based: prescribed | 20 | 0.49 | 0.20 | 20/20 | 0.16 | 20/20 |
| Rule-based: prescribed, reassigned at random | 20 | 0.55 | 0.19 | 20/20 | 0.16 | 20/20 |
| Rule-based: prescribed, reassigned, history wiped | 20 | 0.55 | 0.19 | 20/20 | 0.16 | 20/20 |
| Rule-based: acquired from own history | 20 | 0.35 | 0.19 | 20/20 | 0.15 | 20/20 |
| Rule-based: acquired, history wiped | 20 | 0.19 | 0.19 | 8/20 | 0.15 | 15/20 |
| Rule-based: acquired, no communication | 20 | 0.25 | 0.19 | 18/20 | 0.15 | 20/20 |
| Rule-based: label preference, fixed names | 20 | 0.09 | 0.19 | 0/20 | 0.15 | 0/20 |
| Rule-based: label preference, Latin-square shuffle | 20 | 0.09 | 0.19 | 0/20 | 0.15 | 0/20 |
| Rule-based: label preference, balanced shuffle | 20 | 0.10 | 0.19 | 1/20 | 0.15 | 1/20 |

**Exploratory findings** (not in the plan).

- *The label preference moves rather than vanishes.* For each agent we took the label, as that agent sees it, of the colleague it told most. Under the balanced shuffle this was "A1" for 88 of 90 agents; under the Latin-square shuffle, "A1" or "A4" for 50 of 90. Window-to-window partner overlap exceeded its null in 3 of 10 balanced runs and 9 of 10 Latin-square runs. The hub disappears, but label-driven pairings remain, so partner-level conclusions need a baseline that holds label positions fixed.
- *Output.* Median items completed (of about 600 arriving): gpt-4.1-mini 158 identical vs 49 told a specialty (−69%); Haiku 117.5 vs 82 (−30%); rule-based agents 82.5 with no source vs 50 prescribed (−39%). In these assignment conditions, prescribed specialties reduced

collective output; we do not attribute the loss to specialization itself, since fixed assignments also restrict who can take which item.

- *Haiku 4.5 with fixed names* (3 runs, before the plan): hub above baseline in 2 of 3, "A1" top receiver in 2 of 3, a weaker label preference than gpt-4.1-mini's.

**Deviations from the plan:** none. No run failed or was repeated. The 47 runs made for the plan cost $11.93.

### A6. Exploratory LLM conditions

Whatever was prescribed persisted, and we detected no other persistent differentiation: dispositions produced persistent differences in working pace but not in task type, and hidden skill differences were never discovered. These runs (30 September 2026, fixed names, 3 runs per cell unless noted) preceded the analysis plan and are reported as exploratory (`diffenv_results/final/`).

Cells give median R against median baseline [runs above baseline / runs].

*Table 9.* Exploratory LLM conditions

| Condition | Model | Work given opportunity | Interaction | Favored task type | Items done (median) |
|---|---|---|---|---|---|
| C0 rule-based, identical | — | 0.13 vs 0.18 [0/3] | 0.07 vs 0.18 [1/3] | 0.12 vs 0.18 [0/3] | 57 |
| C0+ rule-based, built-in differences | — | 0.80 vs 0.19 [3/3] | 0.40 vs 0.19 [3/3] | 0.34 vs 0.21 [2/3] | 58 |
| C1 identical | Haiku 4.5 | 0.07 vs 0.18 [0/3] | 0.15 vs 0.20 [0/3] | 0.09 vs 0.19 [0/3] | 121 |
| C1 identical | gpt-4.1-mini | 0.10 vs 0.18 [0/3] | 0.20 vs 0.19 [2/3] | 0.11 vs 0.19 [0/3] | 151 |
| C1 + exit option | Haiku 4.5 | 0.08 vs 0.19 [0/3] | 0.15 vs 0.18 [1/3] | 0.12 vs 0.19 [0/3] | 116 |
| C1 + exit option | gpt-4.1-mini | 0.20 vs 0.19 [2/3] | 0.33 vs 0.19 [3/3] | 0.10 vs 0.19 [0/3] | 159 |
| C1 + hidden competence | Haiku 4.5 | 0.11 vs 0.20 [0/3] | 0.07 vs 0.19 [0/3] | 0.08 vs 0.19 [0/3] | 38 |
| C1 + hidden competence | gpt-4.1-mini | 0.10 vs 0.18 [0/3] | 0.29 vs 0.18 [2/3] | 0.08 vs 0.19 [0/3] | 47 |
| C3 disposition | Haiku 4.5 | 0.56 vs 0.19 [3/3] | 0.13 vs 0.19 [0/3] | 0.12 vs 0.19 [0/3] | 125 |
| C3 disposition | gpt-4.1-mini | 0.62 vs 0.19 [3/3] | 0.46 vs 0.19 [3/3] | 0.11 vs 0.19 [0/3] | 156 |

Reading notes. Interaction measures in fixed-name gpt-4.1-mini runs carry the "A1" label preference (A5) and are not interpreted. With the exit option, no agent left in any run. With hidden competence, the share of an agent's work on its best type stayed near chance (about

0.25) in all six runs, and fewer items were completed because failed work does not count. The specialty condition (C2) is reported with the plan-specified cells in A5.

### A7. Empirical reference cases

The same function, unchanged, finds persistent member-level differences in every reference group, each against its own label-permutation baseline (`diffenv/run_substrates.py`, `diffenv/run_external.py`; sources and licenses in `DATA.md`). Rows record different behaviors over different windows, so values are compared with their own baselines, not across rows.

*Table 10.* Empirical reference cases

| Group (source) | Members | Window | Behavior scored per member and window | R | Baseline (95th pct.) |
|---|---|---|---|---|---|
| Clonal raider ants (Alciatore et al. 2021) | 154 | 9 windows | Time within 1 mm of nestmates | 0.68 | 0.13 |
| Same colonies | 154 | 9 windows | Activity / isolation / speed | 0.48 / 0.53 / 0.52 | 0.13 |
| Office workers, 2013 (Génois et al. 2015) | 87 | 10 days | Face-to-face contact slices per day | 0.34 | 0.16 |
| Office workers, 2015 (Génois & Barrat 2018) | 204 | 10 days | Face-to-face contact slices per day | 0.28 | 0.14 |
| Online game players, strangers matched into teams (Cui, Chung & Hanson-Holtry 2017) | 1,501 | Blocks of 5 matches | Share of picks in the "carry" role | 0.51 | 0.15 |
| Same, blocks of 10 matches / calendar quarters | 721 / 715 | — | Same | 0.58 / 0.52 | 0.15 / 0.19 |
| Carpenter ants, first look (Mersch, Crespi & Keller 2013) | 822 | Colony-days | Interactions per day | 0.56 | 0.04 |
| Honeybees, first look (Gernat et al. 2018) | 2,382 | Colony-days | Food exchanges (trophallaxis) per day | 0.43 | 0.18 |
| Laboratory economies with induced comparative advantage (Kimbrough, Smith & Wilson 2008) | 158 | Production days (sessions as groups) | Share of production on the first good | 0.84 | 0.03 |

The laboratory economies are a human positive control: comparative advantage is built into the design, as specialties are in C2. The two first looks are labeled preliminary because their data preparation has not yet been checked as fully as the main cases'. A published LLM-agent simulation in which 100 agents write their own memory (Takata, Masumori & Ikegami 2024; 10 trials per message range) was scored with the same function on logs the authors shared for this analysis. Persistence in movement rate is 0.51 when agents cannot hear one another, 0.49 at the paper's main setting, 0.24 at an intermediate range and 0.11 when every agent hears

every other, against baselines of 0.11–0.12; memory length declines the same way (0.82, 0.61, 0.45, 0.11). The authors have seen these results and agreed to their reporting. The logs are not redistributed and are available from the authors on request.

### A8. Sample provenance report

One command answers the first question, Is there structure?, for any compatible log, with every value against its own baseline: `python3 provenance_report.py --csv examples/example_log.csv` (about 4 seconds, no API key).

**Log format.** One row per action, with a header. `run` identifies an independent run or group; `time` is an integer step; `action` is `work` (a task action) or `tell` (a directed interaction), and other actions are ignored; `partner` is the recipient of a `tell`; `task_type` is the type of the task worked. Optional rows with action `opp` mark the steps at which a member had something to work on, which enables the opportunity-adjusted individual measure. Rows taken from the example file:

```
run,time,member,action,partner,task_type,opportunity
fixed_names_seed0,0,A3,work,,gamma,
fixed_names_seed0,0,A4,work,,beta,
fixed_names_seed0,0,A3,opp,,,1
fixed_names_seed0,1,A3,tell,A7,,
fixed_names_seed0,1,A4,tell,A7,,
```

**Output** for three stored fixed-name gpt-4.1-mini runs (`examples/example_report.md`). Cells are observed persistence / baseline; **bold** is above baseline.

*Table 11.* Sample provenance report for three stored runs

| Run | Individual | Functional | Relational | Structural | Top receiver |
|---|---|---|---|---|---|
| fixed_names_seed0 | 0.12 / 0.19 | 0.13 / 0.18 | 0.36 / 0.39 | **0.38** / 0.18 | A1 |
| fixed_names_seed1 | 0.10 / 0.18 | 0.04 / 0.19 | 0.36 / 0.37 | **0.44** / 0.19 | A4 |
| fixed_names_seed2 | 0.10 / 0.19 | 0.11 / 0.19 | 0.33 / 0.35 | **0.35** / 0.19 | A1 |

Across runs, only the structural dimension is above baseline (3 of 3; median 0.38 [0.35, 0.44] against 0.19). The report's signature line flags the same member as top receiver in 2 of 3 runs, with a mean cross-run rank correlation of interactions received of +0.75. That recurrence points to prescribed or label-induced structure, which the balanced-shuffle condition then confirms (A5). The report ends by naming the matched conditions needed for the other two questions, Is it robust? and Where did it come from?, since one condition's logs establish persistent differentiation, not its source.

### A9. Reproduction

Every synthetic and known-answer number in the proposal and in A4–A6 and A8 was reproduced exactly from a fresh copy of the repository on a clean machine, with no API key and no network access beyond installing packages. The reference-case values (A7) need the original data deposits and were not part of this check.

**Clean-environment checks (17 and 19 October 2026).** On 17 October the repository's files were copied to a new Linux machine (Python 3.13, a new virtual environment, `pip install -r requirements.txt`: numpy 2.5, matplotlib 3.11, openpyxl 3.1), whereas the results were produced on macOS with Python 3.9 and numpy 2.2. Every step in the table except the functional-baseline calibration, which was added on 19 October, passed or matched the stored results. On 19 October the whole check was repeated from a fresh copy of the repository as submitted, in a new virtual environment (Python 3.13.16, numpy 2.5.4, matplotlib 3.11.2), now including the calibration script; every step again passed or matched the stored results exactly. Times are from the 19 October run.

*Table 12.* Reproduction steps and results of the clean-environment check

| Step | Command (from `code/`) | Time | Result |
|---|---|---|---|
| Estimator fixtures | `python3 -m diffenv.test_measures` | 38 s | All fixtures pass |
| Label-shuffle checks | `python3 -m diffenv.test_alias` | < 1 s | All pass; balanced labels spread tells over all 9 agents, most-told 11% |
| Pipeline (mock model, budget caps, key handling) | `python3 -m diffenv.test_pipeline` | < 1 s | All pass |
| Sample provenance report | `python3 provenance_report.py --csv examples/example_log.csv` | 4 s | Identical to `examples/example_report.md` |
| Plan-specified comparisons | `python3 score_plan.py` | 54 s | Identical to A5 |
| Sensitivity analyses | `python3 score_plan_sensitivity.py` | 55 s | Identical to A5 |
| Functional-baseline calibration | `python3 functional_null_sensitivity.py` | 5.5 min | Identical to the A5 calibration table (19 October check only) |
| Known-answer test, 10 conditions × 20 runs | `python3 -m diffenv.ground_truth --seeds 20 --out results/gt` | 8.5 min | All 1,600 run-level values and all three contrasts identical to the stored results |
| Rule-agent floor (C0) | `python3 -m diffenv.run --condition c0 --seeds 0 1 2 --visibility private` | 6 s | Runs; 0 API calls |
| Re-score stored LLM runs | `python3 -m diffenv.report diffenv_results/final/specimens` | < 1 s | Identical to A6 |
| Figure 1 | `cd figures && python3 make_figure1_v4.py` | 3 s | Redrawn from stored results |

**New LLM runs** need API keys in `keys.env` at the repository root (never committed or printed). Each run is 900 short calls with hard caps on calls and cost, and an interrupted run resumes from its journal without repeating a call. Cost per run: about $0.16 on gpt-4.1-mini and $0.43–$0.48 on Haiku 4.5. The scripts that produced the plan-specified runs are `run_replications.sh` and `run_balanced.sh`.

**Stored runs.** Each LLM run is one JSON specimen holding the configuration, prompts, the full event log, every raw model reply with token counts and latency, and cost. Model versions are recorded per reply. The repository holds the runs reported in the proposal and this appendix, 81 LLM runs ($22.06) and 6 rule-based control runs, in code/diffenv_results/, and archives separately, in archive/excluded_runs/, the nine further LLM runs made in Stage 1 but not reported (table below).

**Runs made but not reported.** Stage 1 made 90 LLM runs, all on gpt-4.1-mini or Haiku 4.5, at a recorded cost of $22.58. The nine not reported are listed here so that every run is disclosed; each file is in `archive/excluded_runs/` with an inventory.

*Table 13.* LLM runs made in Stage 1 but not reported

| Runs | Date | Calls per run | Cost | Purpose | Why not reported |
|---|---|---|---|---|---|
| 1 connectivity check (gpt-4.1-mini, identical agents) | 29 Sep | 3 | $0.0004 | API and logging check | Smoke test |
| 5 pilots (gpt-4.1-mini; identical agents ×4, specialty ×1) | 29 Sep | 45 | $0.036 in total | Prompt and reply-parser checks; comparison of visibility settings that led to the private-visibility design | Five-round pilots before the reported design was fixed |
| 3 full runs (gpt-4.1-mini, identical agents with an exit option, Latin-square labels) | 30 Sep | 900 | $0.49 in total | Exit option under shuffled labels | Superseded condition: the exit option is reported only with fixed names (A6), and the Latin-square shuffle was replaced by the balanced one |

**Reference data** are not redistributed; `fetch_data.py` and the loaders retrieve them from their original deposits, and `DATA.md` lists sources and licenses. The Takata et al. logs are available from their authors.

**Use of AI.** AI was used as a coding and editing assistant (Claude, Anthropic) in developing the analysis code, the figure script, and the text of the proposal and this appendix. The authors designed the studies and the analysis plan, verified all code and reported values, including the clean-environment reproductions, and are responsible for the content. gpt-4.1-mini and Claude Haiku 4.5 appear in the simulations as objects of study, not as assistants, and are documented in A3.

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